

Intent-Based Networking Engine for an ISP



Networks and Systems Management — GRS Project

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What is my project?

An Intent-Based Networking (IBN) Engine

A Python system that reads a YAML file describing the desired network state and automatically provisions the entire infrastructure, routing protocols, DNS, load balancing, and SDN traffic engineering, without any manual configuration.

Core

- YAML intent → Jinja2 → Netmiko
- OSPF internal + BGP external routing with Quagga
- BIND9 DNS auto-provisioned
- Nginx load balancer (4 replicas)
- Nagios closed-loop monitoring

SDN

- Ryu SDN controller + Open vSwitch
- OpenFlow primary & backup paths
- Engine polls Nagios → detects congestion
- Pushes flow rules via Ryu REST API
- Demo: `tc` latency injection

Key Principle

- Declarative, not imperative
- Engine computes the delta
- Applies only what changed
- Idempotent and repeatable

Motivation



Manual configuration is error-prone

Engineers configure routers, DNS, and routing protocols by hand. Only one typo can bring down a network segment. There is no safety net.



No single source of truth

Network state is scattered across scripts, wikis, and people's memory. There is no declarative representation of what the network should look like.



Slow reaction to network events

When a link degrades, human intervention or slow OSPF convergence is the only response. Traffic engineering is reactive, not proactive.

Technologies



Python 3

IBN engine, Jinja2 templates, Netmiko SSH automation, Ryu REST integration



Docker Compose

Container orchestration — routers, servers, DNS, clients, SDN controller



Quagga / FRR

OSPF (intra-AS link-state) and BGP (inter-AS path-vector) routing daemons



BIND9 DNS

Authoritative DNS; zone files auto-generated by IBN engine from YAML intent



Nginx

Web server (4 replicas) and HTTP load balancer; upstream config auto-generated



Nagios Core

Active monitoring: latency, BGP sessions, availability; drives closed-loop logic



Ryu + OpenFlow

SDN controller managing OVS flow tables; enables sub-second traffic rerouting



Open vSwitch

Programmable virtual switch in router containers; responds to Ryu flow rules

Architecture

